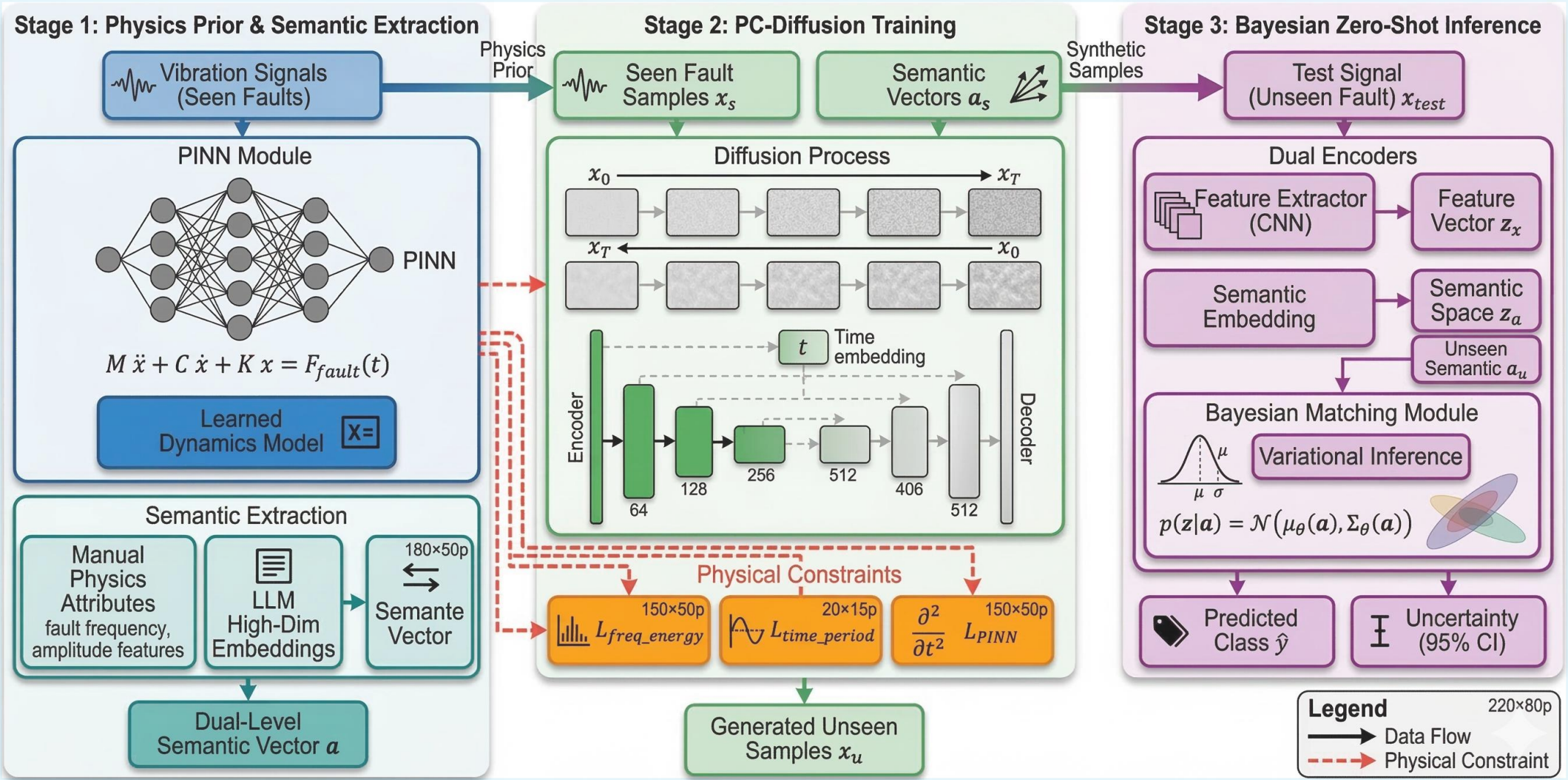
Stage 1: Physics Prior & Semantic Extraction
Vibration Signals (Seen Faults)
PINN Module
PINN
$M\ddot{x} + C\dot{x} + Kx = F_{fault}(t)$
Learned Dynamics Model
Semantic Extraction
Manual Physics Attributes
fault frequency, amplitude features
LLM High-Dim Embeddings
180×50p
Semante Vector
Dual-Level Semantic Vector a
Physics Prior
Stage 2: PC-Diffusion Training
Seen Fault Samples x_s
Semantic Vectors a_s
Diffusion Process
x_0
x_T
Time embedding
Encoder
Decoder
64
128
256
512
406
512
Physical Constraints
150×50p
L_{freq_energy}
20×15p
L_{time_period}
150×50p
$\frac{\partial^2}{\partial t^2}$ L_{PINN}
Generated Unseen Samples x_u
Synthetic Samples
Stage 3: Bayesian Zero-Shot Inference
Test Signal (Unseen Fault) x_{test}
Dual Encoders
Feature Extractor (CNN)
Feature Vector z_x
Semantic Embedding
Semantic Space z_a
Unseen Semantic a_u
Bayesian Matching Module
Variational Inference
$p(z|a) = \mathcal{N}(\mu_\theta(a), \Sigma_\theta(a))$
Predicted Class $\hat{y}$
Uncertainty (95% CI)
Legend
220×80p
Data Flow
Physical Constraint